

Environmental Monitoring

Your smartphone can be a handy tool for monitoring your surroundings. From measuring sound levels to checking light intensity, the built-in sensors make it easy to stay informed about your environment. With a bit of creativity, you can even turn it into a DIY weather station!

Ready to get started? Follow these steps:

1. Open your phone's light sensor or sound recorder to measure brightness or noise levels.
2. Place your phone in a steady position to ensure consistent readings.
3. Pair with external hardware like air quality monitors via Bluetooth for more advanced metrics.
4. Process the collected data locally on your phone and visualise it using graphs or tables.
5. Use your insights to make adjustments (e.g., soundproofing a room or optimising lighting).



High-Speed Video Analysis

Ever wondered how athletes perfect their movements or how scientists study motion? Your phone's high-speed camera is the answer! By shooting in slow motion, you can break down fast actions frame-by-frame to uncover insights in sports, experiments, or everyday life.

Here's how to do it yourself:

1. Open your phone's camera and switch to slow-motion mode (e.g., 240 fps or higher).
2. Set up your scene – good lighting and a steady surface make a big difference.
3. Record your subject in motion, ensuring smooth and consistent movement.
4. Open the video in your gallery and use the scrubber to analyse each frame.
5. Take notes or measurements, like time intervals or distances, directly from the footage.



Local Coding Sandbox

Did you know your smartphone can double as a lightweight coding environment? Whether you're testing Python scripts, building web pages, or experimenting with shell commands, your phone's processing power is enough to handle basic development tasks – all without a computer.

Ready to dive in? Here's how to set it up:

1. Download a terminal emulator or coding app like Termux for Android or Play.js for iOS.
2. Set up your coding environment by installing the necessary tools (e.g., Python, Node.js).
3. Write or copy your code into the editor directly on your phone.
4. Run the code using the app's built-in execution capabilities.
5. Debug or modify your scripts, and optionally share your work through local file storage or Git.

